

Physics Research Notes Template

Your Name

April 14, 2026

Contents

Chapter 1 Proton Decay 的 Bayesian 计算	5
1.1 试运行	5
Appendix A Supplementary Calculations	9

Chapter 1

Proton Decay 的 Bayesian 计算

考虑形变势能:

$$V(r, \theta) = V_N(r, \theta) + V_c(r, \theta) + V_{so}(r, \theta) + \frac{\hbar^2}{2\mu r^2} \left(l + \frac{1}{2}\right)^2 \quad (1.1)$$

$$V_N(r, \theta) = -\frac{V_{N0}}{1 + \exp\{r - R_N(\theta)/a_N(\theta)\}} \quad (1.2)$$

$$R_N(\theta) = R_{N0}(1 + \beta_2 Y_{20}(\theta) + \beta_4 Y_{40}(\theta)) \quad (1.3)$$

$$a_N(\theta) = a_{N0} \sqrt{1 + \left(\frac{1}{R_N(\theta)} \frac{dR_N(\theta)}{d\theta}\right)^2} \times [1 + \tilde{\beta}_2 Y_{20}(\theta) + \tilde{\beta}_4 Y_{40}(\theta)] \quad (1.4)$$

$$V_{so} = V_{so0} \lambda_\pi^2 \frac{1}{r} \frac{d}{dr} \frac{1}{1 + \exp[(r - R_{so})/a_{so}]} \vec{\sigma} \cdot \vec{l} \quad (1.5)$$

$$\lambda_\pi^2 = 2fm^2 \quad (1.6)$$

$$V_c(r) = \begin{cases} \frac{Z_D Z_p e^2}{2R_{c0}} \frac{3R_{c0}^2 - r^2}{R_{c0}^2}, & r \leq R_{c0} \\ \frac{Z_D Z_p e^2}{r}, & r > R_{c0} \end{cases} \quad (1.7)$$

1.1 试运行

使用参数 (145Tm, 形变势能,wkb 积分)

```
#System
e2=1.43997 ; hbarc=197.3269718 ; amu=931.49432
zp=69 ; Ap=145 ; mass_excess_p=-27.58 ; mp=Ap*amu+mass_excess_p
z1=68 ; A1=144 ; mass_excess_1=-36.61 ; m1=A1*amu+mass_excess_1 #1-> daughter
z2=1 ; A2=1 ; mass_excess_2=7.288971064 ; m2=A2*amu+mass_excess_2
Q=mp-m1-m2
mu=m1*m2/(m1+m2)
P0=0.580
#Potential
```

```

#Deformation
beta2 = 0.231 # quadrupole
beta4 = -0.068 # hexadecapole

#Polarizaton
beta2tilde = 0
beta4tilde = 0

#spin-orbit
Vso0=6.2 ; rso=1.01*(A1)**(1/3) ; aso=0.75 ; lambda_pi=np.sqrt(2)
L=5; S=0.5; J=5.5

#Vn
a0=0.75 ; r0=1.27*A1**(1/3)-0.1 ; V0=55

#Vc
Rc0 = 1.21 * A1**(1/3)

# experiment data
y=np.array([3.16e-6]) # s
sigma=np.array([0.16e-6])

#a = v0,a0,r0,p0,rc0,beta2,beta4,vso0,rso,aso
def log_prior(a):
    v0,a0,r0,p0,rc0,beta2,beta4,vso0,rso,aso=a
    R=1

# Hyperparameters
v0_mean=55; a0_mean=0.75; r0_mean=6.556; p0_mean=0.580
rc0_mean=6.342; beta2_mean=0.231; beta4_mean=-0.068
vso0_mean=6.2; rso_mean=5.294; aso_mean=0.75

sigma_v0=1*R; sigma_a0=0.05*R; sigma_r0=0.1*R; sigma_p0=0.05*R
sigma_rc0=0.1*R; sigma_beta2=0.05*R; sigma_beta4=0.01*R
sigma_vso0=1*R; sigma_rso=0.1*R; sigma_aso=0.05*R

with multiprocessing.Pool() as pool:
    sampler = emcee.EnsembleSampler(nwalkers, M, log_posterior, args=[y,sigma],a=1.0,
    pool=pool)
    state = sampler.run_mcmc(initial_positions, 200)
    sampler.reset()

```

```
sampler.run_mcmc(state,10000)
```

获得 ^{145}Tm 的 Posterior 分布如下图所示:

(说明之前的 bug 出自于对 model 函数的解包错误, 而非采样参数错误)

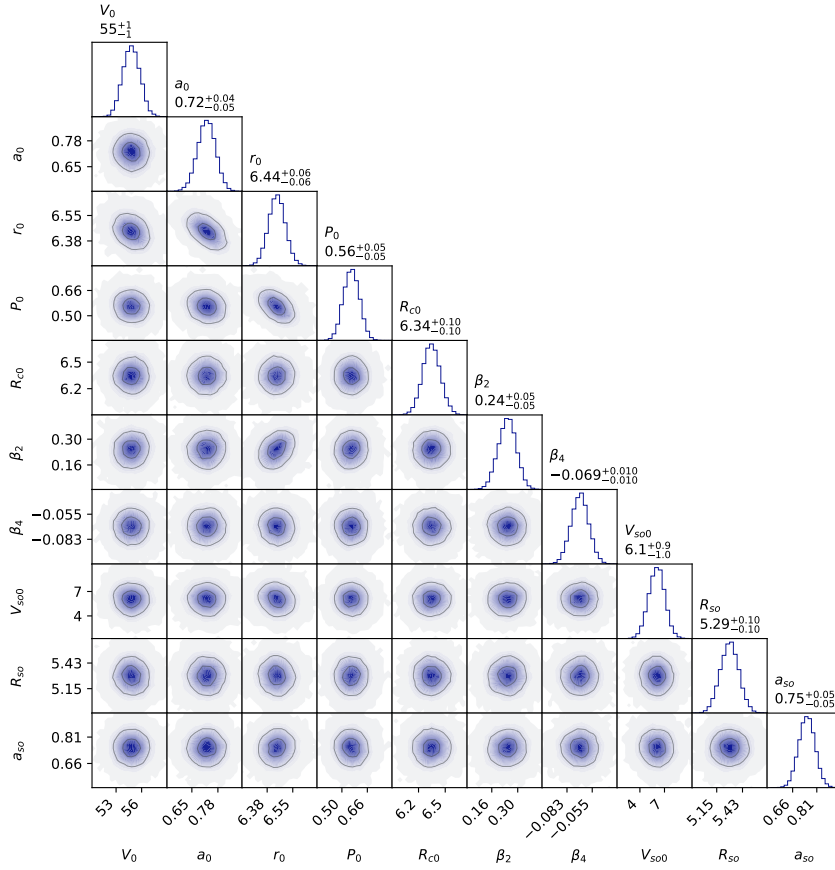


Figure 1.1: Posterior distribution of the parameters for ^{145}Tm proton decay.

上图在 helium1 上计算, 耗时约 3.5d, 速度较慢

后续预计先调整图片样式, 然后做好代码优化加速, 最后可以选取具体核区正式计算

Appendix A

Supplementary Calculations

Put detailed derivations that are too long for the main text here.